

FORT NELSON, BC, Canada

WMO: 719450

Lat: 58.8415N

Lon: 122.5741W

Elev: 382

StdP: 96.82

Time Zone: -7.00 (W07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-36.4	-32.8	-40.6	0.1	-36.3	-37.2	0.1	-32.7	7.2	-9.1	5.9	-12.8	0.5	210	0.646

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.3	28.0	16.8	26.2	16.2	24.5	15.4	18.3	25.1	17.3	23.7	16.4	22.4	2.7	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
16.0	11.9	20.4	14.9	11.1	19.3	14.0	10.4	18.5	53.4	25.0	50.1	23.8	47.3	22.5	23.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.4	6.0	5.2	-38.9	31.2	3.6	1.5	-41.5	32.2	-43.6	33.1	-45.6	34.0	-48.2	35.1		
(5)			-39.0	20.2	3.6	1.3	-41.5	21.1	-43.6	21.9	-45.6	22.6	-48.2	23.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	-0.1	-19.6	-14.4	-8.2	2.8	10.1	15.0	17.3	15.3	9.5	1.1	-11.8	-18.5	(6)
(7)		DBStd	14.41	8.13	7.04	8.08	5.27	4.58	3.30	2.87	3.50	3.93	4.66	6.55	7.03	(7)
(8)		HDD10.0	4318	917	684	564	218	54	4	1	4	55	279	654	883	(8)
(9)		HDD18.3	6753	1176	918	822	466	258	107	54	106	266	536	904	1141	(9)
(10)		CDD10.0	646	0	0	0	2	56	154	226	167	39	2	0	0	(10)
(11)		CDD18.3	40	0	0	0	0	1	7	21	11	0	0	0	0	(11)
(12)		CDH23.3	611	0	0	0	1	39	141	265	152	12	1	0	0	(12)
(13)		CDH26.7	111	0	0	0	0	4	24	54	27	1	0	0	0	(13)
(14)	Wind	WSAvg	1.8	1.2	1.5	2.0	2.4	2.5	2.3	2.1	1.9	1.9	1.7	1.3	1.2	(14)
(15)	Precipitation	PrecAvg	469	24	17	21	20	47	71	83	70	42	32	26	20	(15)
(16)		PrecMax	752	75	40	64	71	130	189	203	205	121	115	82	50	(16)
(17)		PrecMin	321	4	4	2	1	7	20	19	13	1	1	2	0	(17)
(18)		PrecStd	88	14	9	13	17	28	39	38	42	27	21	15	12	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.2	4.7	12.7	20.3	27.2	29.2	30.5	29.5	25.2	17.7	6.2	3.2	(19)
(20)			MCWB	1.2	1.0	5.5	9.9	13.8	16.5	18.4	18.4	15.7	10.1	1.7	0.0	(20)
(21)		2%	DB	-1.7	1.1	9.3	16.7	24.2	26.9	28.2	27.0	21.6	13.4	2.0	-2.2	(21)
(22)			MCWB	-3.1	-1.7	3.3	7.8	12.8	15.7	17.4	17.2	13.9	7.1	-0.5	-3.5	(22)
(23)		5%	DB	-6.1	-1.9	6.5	14.2	21.9	24.8	26.3	24.9	19.2	10.3	-1.0	-5.6	(23)
(24)			MCWB	-6.7	-3.3	1.5	6.2	11.7	14.7	16.8	16.3	12.0	5.6	-2.1	-6.2	(24)
(25)		10%	DB	-9.3	-4.4	3.7	12.0	19.6	22.8	24.4	22.9	17.1	7.8	-3.3	-9.2	(25)
(26)			MCWB	-9.8	-5.4	-0.2	4.9	10.8	13.9	16.0	15.4	11.0	4.4	-4.0	-9.7	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.0	1.4	5.6	10.3	15.5	17.9	20.1	19.7	16.5	10.8	2.1	0.4	(27)
(28)			MCDWB	3.9	4.6	12.2	19.5	23.9	25.6	27.0	26.4	23.2	16.0	5.4	2.8	(28)
(29)		2%	WB	-3.1	-1.5	3.4	8.2	13.7	16.6	18.6	18.2	14.7	7.9	-0.3	-3.5	(29)
(30)			MCDWB	-1.8	0.8	8.9	16.0	22.1	23.9	25.6	24.8	20.0	12.4	1.6	-2.3	(30)
(31)		5%	WB	-6.7	-3.4	1.7	6.6	12.4	15.7	17.6	17.1	12.9	6.0	-2.1	-6.2	(31)
(32)			MCDWB	-6.0	-1.9	6.2	13.3	20.2	22.6	24.0	23.3	17.5	9.5	-0.9	-5.5	(32)
(33)		10%	WB	-9.7	-5.5	0.0	5.3	11.1	14.8	16.7	16.1	11.6	4.5	-4.1	-9.6	(33)
(34)			MCDWB	-9.3	-4.3	3.5	11.3	18.6	21.2	22.5	21.6	15.9	7.7	-3.3	-9.2	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	7.3	10.3	12.0	12.0	12.8	11.8	11.3	11.7	11.5	8.2	6.9	6.7	(35)
(36)			MCDBR	11.2	13.6	14.9	16.6	17.4	16.0	15.1	15.4	15.6	13.8	10.0	10.3	(36)
(37)			MCWBR	10.0	11.2	9.8	9.1	8.3	6.9	6.4	7.2	8.5	8.5	7.9	8.9	(37)
(38)		5% WB	MCDBR	11.2	13.2	14.7	15.8	15.6	13.9	13.4	14.1	14.7	13.1	9.6	10.1	(38)
(39)			MCWBR	10.0	10.8	9.8	9.0	7.6	6.4	6.2	7.1	8.5	8.4	7.7	8.8	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.218	0.246	0.274	0.328	0.349	0.360	0.365	0.351	0.311	0.291	0.230	0.196	(40)	
(41)		taud	2.240	2.325	2.306	2.352	2.422	2.418	2.421	2.475	2.589	2.552	2.326	2.126	(41)	
(42)		Ebn at Noon	653	811	880	880	883	875	861	849	841	754	633	538	(42)	
(43)		Edh at Noon	41	64	89	103	103	105	103	91	69	53	38	31	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.50	1.37	2.90	4.64	5.39	5.55	5.37	4.45	2.94	1.46	0.59	0.30	(44)	
(45)		RadStd	0.04	0.11	0.18	0.20	0.39	0.39	0.41	0.39	0.23	0.13	0.07	0.02	(45)	

Historical Trends

		DBAvg	Heating		Cooling		Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only	+0.49	+2.02	+2.64	N/A	N/A	N/A	N/A	-144	N/A	N/A
(47) Regional (2 neighbors)	+0.53	+2.10	+3.01	N/A	N/A	N/A	N/A	-163	N/A	N/A

Nomenclature: See separate page